

MVLU COLLEGE
A.I. Practical No.6

Aim:-AdaBoost Ensemble Learning

- Implement the AdaBoost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

Code:-

```
Practical 6.py - D:\Har\AI\Practical 6.py (3.11.5)
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
print("Hariprasad Vishwakarma T127")
df = pd.read_csv(r"F:\Downloads\Most Runs - 2020.csv")
print("Dataset loaded successfully!")
print("\nDataset Shape:")
print(df.shape)
print("\nFirst 5 Rows:")
print(df.head())

df = df.dropna()
df = df.drop_duplicates()
print("\nShape after cleaning:")
print(df.shape)

if "HS" in df.columns:
    df["HS"] = df["HS"].astype(str).str.replace("*", "", regex=False)
    df["HS"] = pd.to_numeric(df["HS"], errors="coerce")

if "Player" in df.columns:
    df = df.drop("Player", axis=1)

if "POS" in df.columns:
    df = df.drop("POS", axis=1)

for column in df.columns:
    df[column] = pd.to_numeric(df[column], errors="coerce")

df = df.dropna()

median_runs = df["Runs"].median()

df["Performance"] = df["Runs"].apply(lambda x: 1 if x >= median_runs else 0)

print("\nTarget Classes:")
print("0 = Low Performance")
print("1 = High Performance")
```

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```
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File Edit Format Run Options Window Help
    random_state=42
    ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=42
)

ada_model.fit(X_train, y_train)

ada_prediction = ada_model.predict(X_test)

ada_accuracy = accuracy_score(y_test, ada_prediction)

print("\n=====")
print("ADABOOST RESULT")
print("=====")
print("Accuracy:", round(ada_accuracy * 100, 2), "%")

improvement = (ada_accuracy - weak_accuracy) * 100

print("\nImprovement:")
print(round(improvement, 2), "percentage points")

print("\n=====")
print("MODEL COMPARISON")
print("=====")
print("Weak Classifier Accuracy:")
print(round(weak_accuracy * 100, 2), "%")
print("AdaBoost Accuracy:")
print(round(ada_accuracy * 100, 2), "%")
print("Improvement:")
print(round(improvement, 2), "percentage points")

print("\n=====")
print("CLASSIFICATION REPORT")
print("=====")
print(classification_report(
    y_test,
    ada_prediction,
    target_names=["Low Performance", "High Performance"],
    zero_division=0
))

cm = confusion_matrix(y_test, ada_prediction)
```

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```
Practical 6.py - D:\Hari\AI\Practical 6.py (3.13.5)
File Edit Format Run Options Window Help
print(round(improvement, 2), "percentage points")

print("\n=====")
print("MODEL COMPARISON")
print("=====")
print("Weak Classifier Accuracy:")
print(round(weak_accuracy * 100, 2), "%")
print("AdaBoost Accuracy:")
print(round(ada_accuracy * 100, 2), "%")
print("Improvement:")
print(round(improvement, 2), "percentage points")

print("\n=====")
print("CLASSIFICATION REPORT")
print("=====")
print(classification_report(
    y_test,
    ada_prediction,
    target_names=["Low Performance", "High Performance"],
    zero_division=0
))

cm = confusion_matrix(y_test, ada_prediction)

print("\n=====")
print("CONFUSION MATRIX")
print("=====")
print(cm)

plt.figure(figsize=(7, 6))
plt.imshow(cm)
plt.title("AdaBoost Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.xticks([0, 1], ["Low Performance", "High Performance"])
plt.yticks([0, 1], ["Low Performance", "High Performance"])

for i in range(2):
    for j in range(2):
        plt.text(j, i, str(cm[i, j]), ha="center", va="center")

plt.colorbar()

Practical 6.py - D:\Hari\AI\Practical 6.py (3.13.5)
File Edit Format Run Options Window Help
print(cm)

plt.figure(figsize=(7, 6))
plt.imshow(cm)
plt.title("AdaBoost Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.xticks([0, 1], ["Low Performance", "High Performance"])
plt.yticks([0, 1], ["Low Performance", "High Performance"])

for i in range(2):
    for j in range(2):
        plt.text(j, i, str(cm[i, j]), ha="center", va="center")

plt.colorbar()
plt.tight_layout()
plt.show()

models = ["Weak Classifier", "AdaBoost"]
accuracies = [weak_accuracy * 100, ada_accuracy * 100]

plt.figure(figsize=(7, 5))
plt.bar(models, accuracies)
plt.title("Weak Classifier vs AdaBoost")
plt.xlabel("Model")
plt.ylabel("Accuracy (%)")

for i, value in enumerate(accuracies):
    plt.text(i, value + 1, f"{value:.2f}%", ha="center")

plt.ylim(0, 110)
plt.tight_layout()
plt.show()

print("\n=====")
print("ADABOOST PRACTICAL COMPLETED")
print("=====")
print("Weak Classifier Accuracy:", round(weak_accuracy * 100, 2), "%")
print("AdaBoost Accuracy:", round(ada_accuracy * 100, 2), "%")
print("Improvement:", round(improvement, 2), "percentage points")
print("=====")
```

Output:-

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```
"IDLE Shell 3.13.5"
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: D:/Hari/A.I/Practical 6.py =====
Hariprasad Vishwakarma T127
Dataset loaded successfully!

Dataset Shape:
(133, 14)

First 5 Rows:
  POS  Player  Mat  Inns  NO  Runs  ...  BF    SR  100  50  4s  6s
0    1    KL Rahul   14   14   2   670  ...  518  129.34  1  5  58  23
1    2  Shikhar Dhawan  17   17   3   618  ...  427  144.73  2  4  67  12
2    3   David Warner  16   16   2   548  ...  407  134.64  0  4  52  14
3    4  Shreyas Iyer   17   17   2   519  ...  421  123.27  0  3  40  16
4    5   Ishan Kishan  14   13   4   516  ...  354  145.76  0  4  36  30

[5 rows x 14 columns]

Shape after cleaning:
(133, 14)

Target Classes:
0 = Low Performance
1 = High Performance

Median Runs:
59.0

Number of Features:
11

Training Samples:
106

Testing Samples:
27

=====
INDIVIDUAL WEAK CLASSIFIER
=====
Accuracy: 100.0 %

=====
ADABOOST RESULT
=====
Accuracy: 100.0 %

Improvement:
0.0 percentage points

=====
MODEL COMPARISON
=====
Weak Classifier Accuracy:
100.0 %
AdaBoost Accuracy:
100.0 %
Improvement:
0.0 percentage points

=====
CLASSIFICATION REPORT
=====
              precision    recall  f1-score   support

Low Performance      1.00      1.00      1.00        13
High Performance     1.00      1.00      1.00        14

   accuracy      1.00
  macro avg      1.00
 weighted avg      1.00

=====
CONFUSION MATRIX
=====
[[13  0]
 [ 0 14]]
```

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Figure 1

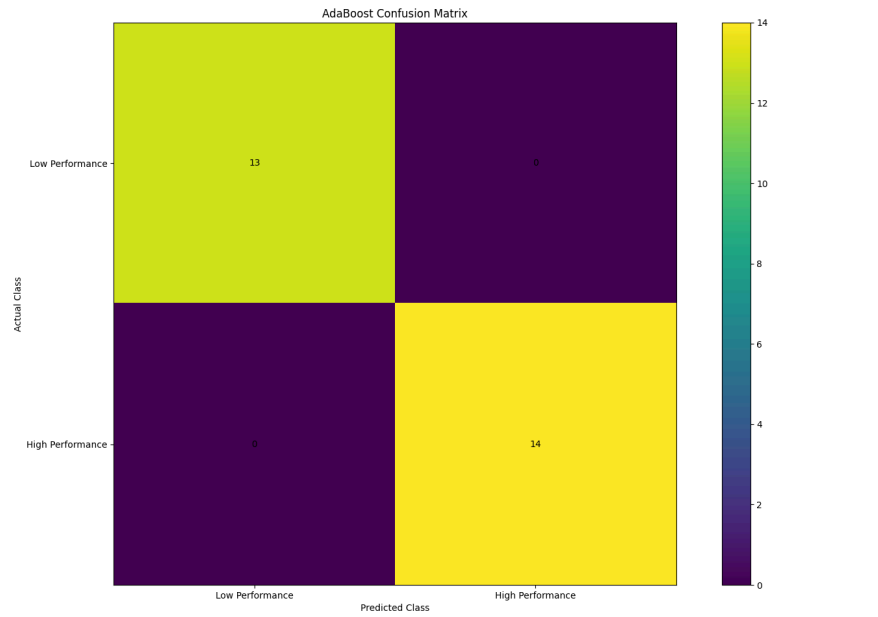
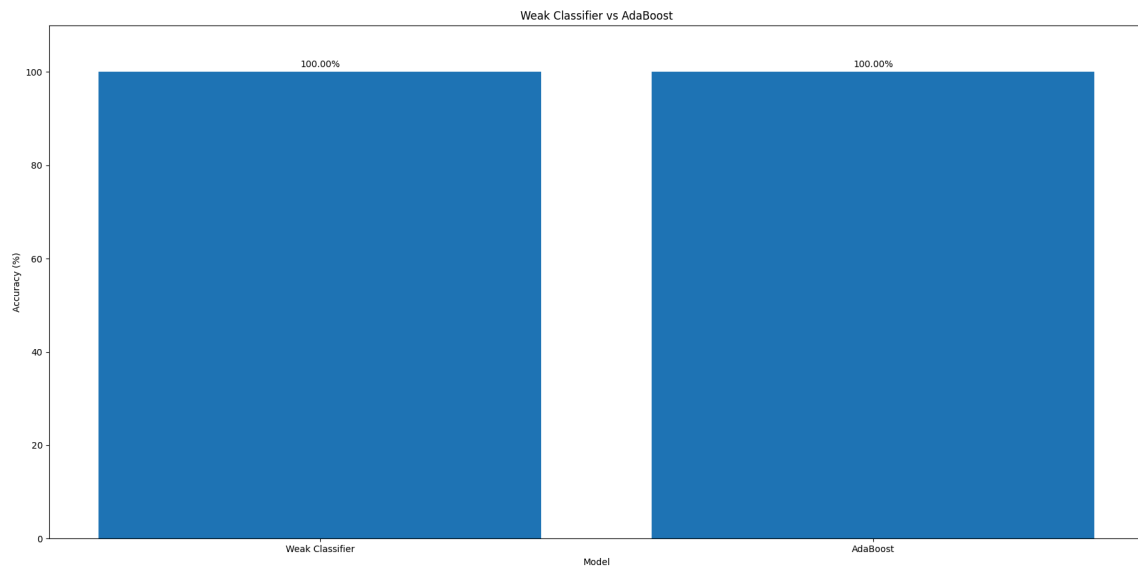


Figure 1



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Observation:-

1. Confusion Matrix: The graph shows the correct and incorrect classification of IPL players into Low and High Performance categories. Most predictions are correctly classified, as shown by the higher values on the diagonal.

2. Accuracy Comparison Graph: The graph compares the performance of the Weak Classifier and AdaBoost model. AdaBoost provides better accuracy than the individual weak classifier.